Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 2 (Generated practice paper - Set 03)

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
TOTAL	

Do not write outside the box

Answer all questions in the spaces provided.

0 1 . 1 A student measured the pH of a stream with a meter that reads to 0.01 pH units but had not been calibrated. Which statement is correct? Tick **one** box.

[1 mark]

A The readings are precise but may not be accurate ☐

B The readings are accurate but not precise ☐

C The readings are both accurate and precise ☐

D The readings are reliable because the meter is electronic ☐

0 1 . 2 Explain what is meant by an **anomalous result** and how a student should decide whether to include it.

[2 marks]

.....

.....

.....

.....

0 1 . 3 Explain why a preliminary study is carried out before the main investigation.

[2 marks]

.....

.....

.....

.....

5

0 2 . 1 State the Brundtland Commission definition of sustainable development.

[1 mark]

.....

.....

0 2 . 2 Explain the difference between a negative feedback mechanism and a positive feedback mechanism, giving **one** example of each in the climate system.

[4 marks]

.....

.....

.....

Do not write outside the box

Do not write outside the box

0 2 . 3 Explain why a diverse ecosystem is more likely to resist change than an agroecosystem.

[1 mark]

0 2 . 4 Evaluate the extent to which copying the principles of natural systems could make human activities more sustainable.

[9 marks]

Do not write outside the box

Extra space

15

0 3 . 1 Which of the following is an example of habitat destruction rather than a change in abiotic factors? Tick one box.

[1 mark]

A Thermal pollution of a river by a power station ☐

B Flooding a valley to create a reservoir ☐

C Lowering the water table by groundwater abstraction ☐

D Acid mine drainage entering a stream ☐

0 3 . 2 Give **four** human activities that cause complete habitat destruction.

[4 marks]

1 .....

.....

2 .....

.....

3 .....

.....

4 .....

.....

5

0 4 . 1 Explain how the relationship between coral polyps and zooxanthellae allows a high biomass of life in nutrient-poor tropical seas.

[4 marks]

.....

.....

.....

.....

.....

.....

.....

.....

.....

Do not write outside the box

0 4 . 2 Explain why growing coral reefs help to resist climate change.

[2 marks]

0 4 . 3 Discuss the importance of tropical coral reefs and evaluate the methods used to conserve them.

[9 marks]

Do not write outside the box

Do not write outside the box

Extra space

15

0 5 Table 1 shows features of some fishing methods.

0 5 . 1 Complete Table 1.

[6 marks]

Table 1

Method	Target species	One environmental disadvantage	
Demersal trawling			
Purse seining			
Pelagic long-lining			

0 5 . 2 Explain why by-catch is a particular problem with shrimp trawling and drift netting.

[2 marks]

Do not write outside the

0 5 . 3 Explain what is meant by **ghost fishing** and describe **two** ways it can be reduced.

[2 marks]

.....

.....

.....

.....

10

0 6 . 1 Give **two** natural processes that are driven by solar energy.

[2 marks]

1 .....

.....

2 .....

.....

0 6 . 2 Explain why natural processes can take place at low temperatures.

[2 marks]

.....

.....

.....

.....

0 6 . 3 Give **one** human activity that uses high temperatures to achieve a result that natural systems achieve at low temperature.

[1 mark]

.....

.....

5

Do not write outside the box

0 7 . 1 Give **two** habitats created by farming that support important wildlife communities.

[2 marks]

1 .....

.....

2 .....

.....

0 7 . 2 Explain how drainage and nutrient enrichment of farmland reduce biodiversity.

[2 marks]

.....

.....

.....

.....

0 7 . 3 Explain **one** problem that can be caused by introducing a biological control species.

[1 mark]

.....

.....

5

0 8 A new pond was dug in a nature reserve. Table 1 shows the vegetation recorded at intervals over 60 years.

Table 1

Years after creation	Dominant vegetation	Water depth at centre / m
1	Single-celled algae, floating duckweed	2.0
5	Submerged pondweeds; reeds at margins	1.8
15	Reeds and reedmace over most of pond; water lilies	1.0
30	Marsh plants, sedges; willow and alder at edges	0.2
60	Willow / alder carr; oak seedlings	0 (soil waterlogged)

0 8 . 1 Name the type of sere shown in Table 1 and identify the pioneer species.

[2 marks]

Do not write outside the box

Do not write outside the box

0 8 . 2 Explain the changes shown in Table 1.

[5 marks]

0 8 . 3 Explain why a lithosere and a hydrosere in the same region end with the same climax community.

[2 marks]

0 8 . 4 Describe the early stages of a lithosere.

[1 mark]

10

0 9 Table 1 shows some designated protected areas in the UK.

0 9 . 1 Complete Table 1 to show the legislation under which each area is designated and its key feature.

[4 marks]

Table 1

Designation	Designated under	Key feature	
Site of Special Scientific Interest (SSSI)			
Special Protection Area (SPA)			
Ramsar site			
Marine Conservation Zone (MCZ)			

0 9 . 2 Explain how designating a protected area protects habitats and species.

[3 marks]

0 9 . 3 Explain why voluntary agri-environmental schemes are needed in addition to legislation.

[3 marks]

Do not write outside the box

Do not write outside the box

.....

.....

.....

.....

.....

.....

.....

10

1 0 Table 1 shows the number of adult great crested newts caught in bottle traps in a pond over five years, and the number of ponds in the area in which the species was detected using environmental DNA (eDNA).

Table 1

Year	Adults caught in traps	Ponds with newts detected by eDNA (out of 20)
2019	14	6
2020	22	9
2021	18	11
2022	31	13
2023	27	15

1 0 . 1 Calculate the percentage increase in the number of ponds where newts were detected between 2019 and 2023.

[2 marks]

Answer ..... %

1 0 . 2 Explain how eDNA detects the presence of newts without catching them.

[2 marks]

.....

.....

Do not write outside the box

1 0 . 3 Suggest why trap catches are a less reliable measure of population change than the eDNA survey.

[2 marks]

1 0 . 4 Evaluate the contribution of new technologies to ecological monitoring and wildlife conservation.

[9 marks]

Do not write outside the box

Extra space

15

Write an essay on one of the following topics.

11 . 1 Discuss the methods that can be used to conserve biodiversity and evaluate their effectiveness.

[25 marks]

OR

11 . 2 Explain how the physical features of the Earth and the activities of living organisms have combined to create and maintain conditions suitable for life.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

Do not write outside the box

Do not write outside the box

Do not write outside the box

Do not write outside the box

Do not write outside the box

END OF QUESTIONS